Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech. ***Time: 03 Hours***

Code: 104305 ***Subject: Signals and Systems*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven question only): **[2 x 7 = 14]**

(a) The minimum sampling rate to avoid aliasing for $x(t)= 5\cos(400\pi t)$ is
(i) 100 Hz (ii) 400 Hz
(ii) 800 Hz (iv) $300Hz$

(b) Which of the following system is memory less?
(i) $h(t)=0, t\neq 0$ (ii) $h(t)=x(t-1)$
(iii) $h(t)=0, t=0$ (iv) $h(t)=kx(t-2)$

(c) The ROC of $x(t) = e^{-2t}u(t) + e^{-3t}u(t)$ is
(i) $\sigma > 2$ (ii) $\sigma > 3$
(iii) $\sigma > -3$ (iv) $\sigma > -2$

(d) Time period of: $x(t) = 3\cos(20t + 5) + \sin(8t-3)$ is
(i) $\frac{1}{10}$ sec (ii) $\frac{1}{20}$ sec
(iii) $\frac{2}{5}$ sec (iv) $\frac{2}{15}$ sec

(e) The power of the signal: $x(t) = \cos(t)$.
(i) $\frac{1}{2}$ (ii) $\frac{3}{2}$
(iii) 1 (iv) 2

(f) The Fourier transform of an exponential signal is
(i) $\frac{1}{-a+j\omega t}$ (ii) $\frac{1}{a-j\omega t}$
(iii) $\frac{1}{a+j\omega t}$ (iv) $\frac{1}{-a-j\omega t}$

(g) $Y(t) = x(t/5)$ is _____
(i) Compressed signal (ii) Expanded signal
(iii) Time shifted signal (iv) Amplitude scaled signal by factor 1.5

(h) Which one of the following is an example of a bounded signal ?
(i) $e^{t}\cos(\omega t)$ (ii) $e^{t}\cos(-\omega t)$
(iii) $e^{-t}\cos(\omega t)$ (iv) $e^{t}\sin(\omega t)$

(i) The simplified valve of $X(n)= \sum_{n=-5}^{5}\sin(2n)\,\delta(n+7)$ is
(i) 0 (ii) $sin10$
(ii) $-\sin 10$ (iv) 1

(j) If $X(S)= \frac{4S-1}{S^2+6S-3}$,then initial value of $x(0)$ will be
(i) $\frac{1}{3}$ (ii) $\frac{1}{4}$
(iii) 4 (iv) 3

Q.2 (a) Find the inverse Z-transform of $X(Z)= \frac{1}{(1+3Z^{-1}+2Z^{-2})}$ ROC:$|Z| > 2$.. **[7]**

(b) $y(t)= 12x(t) + 7$. Check whether the system is (i) time variant time-invariant (ii)causal non-causal (iii)linear non-linear **[7]**

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Q.3 (a) Find the even and odd components of the sequence
$X(n)= 5\delta(n+4) + 4\delta(n+3) + 3\delta(n+2) + \delta(n+1)$ [7]

(b) Determine the power of the signal $x(t)= e^{jat}\cos(\omega_o t)$ [7]

Q.4 (a) Find the Fourier transform of $x(t) = \frac{1}{a^2+t^2}$ [6]

(b) find the time response of LTI system with impulse response & input is [8]
$h(t) = 2u(t)-2u(t-3)$
$x(t)= 8ut(t) - 8u(t-5)$.

Q.5 (a) Sketch the signal $x(-4t-3)$ as shown in fig-1 [7]

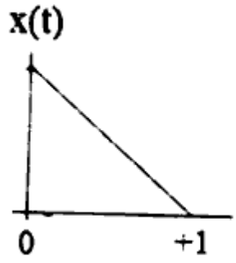


fig. 1

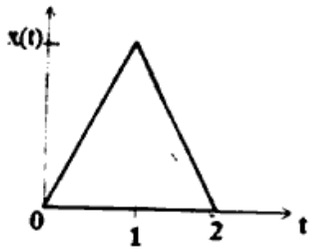


fig. 2

(b) Find the convolution of the following sequence [7]
$x(n)= 2\delta(n+1) - \delta(n) + \delta(n-1) + 3\delta(n-2)$
$h(n)= 3\delta(n-1) + 4\delta(n-2) + 2\delta(n-3)$

Q.6 (a) Compute the DFT of $x(n)= \{0,1,2,4\}$. [7]

(b) Compute the output of the following signals whose impulse response and input are given by $h(t) = e^{-at}u(t)$; $x(t) = e^{at}u(-t)$. $a > 0$ respectively. [7]

Q.7 (a) Find the Laplace Transform of signal as shown in below fig.2[Q.5(a)] [7]

(b) Calculate the fundamental time-period of $x(t)= 1 + \sin\left(\frac{2\pi}{3}t\right)\cos\left(\frac{4\pi}{5}t\right)$. [7]

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Q.8 (a) Determine the Nyquist sampling rate and Nyquist sampling intervals for the signal $x(t)= sinc(100\pi t)\, sinc(200\pi t)$ [7]

(b) Obtain the state transition matrix $\emptyset(t)$ for the system represented by state equation [7]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Q.9 Write short notes on *any two* of the following: [7x2=14]

(a) Comparison between CTFT & DTFT
(b) Aliasing and its effect
(c) Zero-order hold circuit.
(d) Energy & Power signal.

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